

Ques. Switch Statements?

- The match statement is used to perform different actions based on different conditions.
- Instead of writing many **if..else statements**, you can use the match statement.
- its a case sensitive, hello and HELLO is different word.

```
lang = input("What's the programming language you want to learn? ")

match lang:
    case "JavaScript":
        print("You can become a web developer.")

    case "Python":
        print("You can become a Data Scientist")

    case "PHP":
        print("You can become a backend developer")

    case "Solidity":
        print("You can become a Blockchain developer")
```

Output:-

```
What's the programming language you want to learn? PHP
You can become a backend developer
# if value no found then no error
What's the programming language you want to learn? node
The language doesn't matter, what matters is solving problems.
```

Default Value

- Use the **underscore character** `_` as the last case value if you want a code block to execute when there are not other matches:

```
day = 4
match day:
    case 6:
        print("Today is Saturday")
    case 7:
        print("Today is Sunday")
    case _:
        print("Looking forward to the Weekend")
```

Output:-

```
Looking forward to the Weekend
```

Combine Values

- Use the pipe character | as an or operator in the case evaluation to check for more than one value match in one case:

```
day = 4
match day:
    case 1 | 2 | 3 | 4 | 5:
        print("Today is a weekday")
    case 6 | 7:
        print("I love weekends!")
```

Output:-

Today is a weekday

If Statements as Guards

- You can add if statements in the case evaluation as an extra condition-check:

```
month = 5
day = 4
match day:
    case 1 | 2 | 3 | 4 | 5 if month == 4:
        print("A weekday in April")
    case 1 | 2 | 3 | 4 | 5 if month == 5:
        print("A weekday in May")
    case _:
        print("No match")
```

Output:-

A weekday in May

```
subject = input("Enter a subject: ")
score = int(input("Enter a score: "))

match subject:
    # if score is 80 or higher in Physics or Chemistry
    case 'Physics' | 'Chemistry' if score >= 80:
        print("Excellent in Science!")

    # if score is 80 or higher in English or Grammar
    case 'English' | 'Grammar' if score >= 80:
        print("Excellent in English!")

    # if score is 80 or higher in Maths
    case 'Maths' if score >= 80:
        print("Excellent in Maths!")

    case _:
        print(f"Needs improvement in {subject}!")
```

Output:-
Enter a subject: Physics
Enter a score: 2
Needs improvement in Physics!

Switch Statements Using class

```
class Python_Switch:
    def day(self, month):

        default = "Incorrect day"
        return getattr(self, 'case_' + str(month), lambda: default)()

    def case_1(self):
        return "Jan"

    def case_2(self):
        return "Feb"

    def case_3(self):
        return "Mar"

my_switch = Python_Switch()

print(my_switch.day(1))
print(my_switch.day(3))
```

Output:-
Jan
Mar

Switch Statements Using function

```
def number_to_string(argument):
    match argument:
        case 0:
            return "zero"
        case 1:
            return "one"
        case 2:
            return "two"
        case default:
            return "something"

head = number_to_string(2)
print(head)      # Output:- two
```

Using a Dictionary as a Switch

```
def switch_example(value):  
    switch = {  
        1: "One",  
        2: "Two",  
        3: "Three"  
    }  
    return switch.get(value, "Unknown") # Default if key not found  
  
print(switch_example(2)) # Output: Two
```

Dictionary of Functions

```
def case_one():  
    return "One"  
  
def case_two():  
    return "Two"  
  
def case_three():  
    return "Three"  
  
def default_case():  
    return "Unknown"  
  
def switch_example(value):  
    switch = {  
        1: case_one,  
        2: case_two,  
        3: case_three  
    }  
    return switch.get(value, default_case)()  
  
print(switch_example(3)) # Output: Three
```